QM-2
Oct-7

Today: Time-dependent Pert. theory (5.5.1-2, 5.7.1)

<u>Warm-up</u>: Time evolution

Given: Const. Hamiltonian $H_0$

T.I.S.E.: $H_0|n\rangle = E_n|n\rangle$

T.D.S.E: $i\hbar \frac{\partial}{\partial t}|\psi, t\rangle = H_0|\psi, t\rangle$

Initial state: $|\psi, t=0\rangle = |n\rangle$

Show: $|\psi, t\rangle = e^{-iE_n t/\hbar}|n\rangle$

Solves T.D.S.E

$\Rightarrow \quad i\hbar \frac{\partial}{\partial t}|\psi, t\rangle = E_n|\psi, t\rangle$

$$\frac{\partial}{\partial t}|\psi, t\rangle = -\frac{E_n i}{\hbar}|\psi, t\rangle$$

$\Rightarrow |\psi, t\rangle = e^{-\frac{E_n i}{\hbar}t}$

assuming $t_0 = 0$

$$\Rightarrow |\psi, t\rangle = e^{-E_n i (t-0)/\hbar}$$

$$\Rightarrow |\psi, t\rangle = e^{-E_n i t/\hbar} \cdot e^{-E_n i (0)/\hbar}$$

⋮

- General hamiltonian $H(t)$

T.D.S.E : $i\hbar \frac{\partial}{\partial t} |\alpha, t\rangle = H(t) |\alpha, t\rangle$

- time evolution operator : $|\alpha, t\rangle = U(t, t_0) |\alpha, t_0\rangle$
  for any $|\alpha, t_0\rangle$

plugging in T.D.S.E

$$\Rightarrow i\hbar \frac{\partial}{\partial t} U(t, t_0) |\alpha, t_0\rangle = H(t) U(t, t_0) |\alpha, t_0\rangle$$

$$\Rightarrow \boxed{i\hbar \frac{\partial}{\partial t} U(t,t_0) = H(t)\, U(t,t_0)}$$

time evolution operator satisfies this condition

special $H(t) = H_0$

$$i\hbar \frac{\partial}{\partial t} U_0(t,t_0) = H_0 U_0(t,t_0)$$

Solution :

$$U_0(t,t_0) = e^{-iH_0(t-t_0)/\hbar}$$

initial condition : $U(t,t) = \mathbb{1}$

Ex: $|\alpha, 0\rangle = |n\rangle$

$$|\alpha, t\rangle = e^{-iH_0(t-0)/\hbar} |n\rangle$$

$$= e^{-iE_n t/\hbar} |n\rangle$$

## T.D.P.T

- suppose : $H(t) = H_0 + V(t)$
- want to find $U(t,t_0)$ for this hamiltonian

Solve: $i\hbar \frac{\partial}{\partial t} U(t,t_0) = (H_0 + V(t))\, U(t,t_0)$

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power series can be used to find solution.

this equation can be written as an integral equation:

$$U(t,t_0) = U_0(t,t_0) + \frac{1}{i\hbar}\int_{t_0}^{t} dt_1 \, \underbrace{U_0(t,t_1)}_{\text{unperturbed}} V(t_1) \overbrace{U(t_1,t_0)}^{\text{exact}}$$

where, $U_0(t,t_0) = e^{-iH_0(t-t_0)/\hbar}$

ANALOGY:

$$y = 1 + r + r^2 + \ldots\ldots$$

$$= 1 + r(\underbrace{1 + r + r^2 + \ldots\ldots}_{y})$$

$$\boxed{y = 1 + ry}$$

$$= 1 + r + r^2 y$$

$\vdots$

and so on

( keep repeating would kind push y-off to infinitey

this analogy can be applied to the

integral above

$$U(t,t_0) = U_0(t,t_0) + \frac{1}{i\hbar}\int_{t_0}^{t} dt_2\, U_0(t,t_2)\, V(t_2)\Big[\boxed{U_0(t_2,t_0)} + \frac{1}{i\hbar}\int_{t_0}^{t_2} dt_1\, U_0(t_2,t_1)\, V(t_1)\, U(t_1,t_0)\Big]$$

expanding this again.

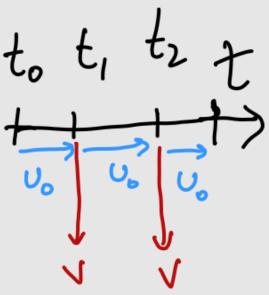
$$= U_0(t,t_0) + \frac{1}{i\hbar}\int_{t_0}^{t} dt_2\, U_0(t,t_2)\, V(t_2)\, U_0(t_2,t_0)$$

$$+ \left(\frac{1}{i\hbar}\right)^2 \int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1\, U_0(t,t_2)\, V(t_2)\, U_0(t_2,t_1)\, V(t_1)\, U_0(t_1,t_0) + O(V^3)$$

$$t \geq t_2 \geq t_1 \geq t_0$$

time evolution

time axis




evolves under unperturbed hamiltonian from $t_0$ to $t_1$

experiences perturbation at $t_1$

$t_1 \to t_2$ again unperturbed hamiltonian

perturbation at $t_2$ again.

this problem can also be solved by another approach:

# Interaction Picture

Define : $|\alpha, t\rangle_I = e^{iH_0 t/\hbar} |\alpha, t\rangle$

Schrödinger ket

Example : $|\psi, t\rangle = e^{-iE_n t/\hbar} |n\rangle$ , $H(t) = H_0$

$$|\psi, t\rangle_I = \underbrace{e^{i\boxed{H_0}t/\hbar}\, e^{-iE_n t/\hbar}}_{\to 1} |n\rangle \qquad H_0 \to E_n$$

$\Rightarrow |\psi, t\rangle_I = |n\rangle$

evolution of $|\alpha, t\rangle_I$ :

$$|\alpha, t\rangle_I = e^{iH_0 t/\hbar} |\alpha, t\rangle = e^{iH_0 t/\hbar}\, U(t, t_0) |\alpha, t_0\rangle$$

$$= \underbrace{e^{iH_0 t/\hbar}\, U(t, t_0)\, e^{-iH_0 t_0/\hbar}}_{U_I(t, t_0)} |\alpha, t\rangle_I$$

plug in $U(t, t_0)$:

$$U_I(t, t_0) = \mathbb{1} + \frac{1}{i\hbar} \int_{t_0}^{t} dt'\, V_I(t')\, U_I(t', t_0)$$

$$= \mathbb{1} + \frac{1}{i\hbar} \int_{t_0}^{t} dt_1\, V_I(t_1) +$$

$$\frac{1}{(i\hbar)^2}\int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1 \; V_I(t_2)\, V_I(t_1) + \dots$$

where,

$$V_I(t) = e^{iH_0 t/\hbar}\, V(t)\, e^{-iH_0 t/\hbar}$$

$$|n\rangle \xrightarrow{\;\mathbb{1}\;} V_I(t')\,|n\rangle$$